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Vidar Bjurström and the Recruitment Team
Blykalla
Stockholm, Sweden

Application: Senior Thermohydraulic Analyst

Dear Vidar Bjurström and Recruitment Team,

I am applying for the Senior Thermohydraulic Analyst position at Blykalla. My background combines conservation-based transient modelling, computational fluid dynamics, heat-transfer and combustion modelling, and scientific programming. I would like to contribute this experience to the development and validation of models supporting Blykalla's fluid systems. The connection between physical modelling, experimental evidence, and engineering decisions is a strong motivation for my application.

At Joyson Safety Systems, I developed Python-based 0-D models of pyrotechnical initiators from mass and energy conservation, combustion, heat transfer, and nozzle-flow relations. The models supplied transient pressure, temperature, mass-flow, and energy-flow conditions for coupled 3-D OpenFOAM simulations of airbag inflators. I compared predicted pressure histories with closed-bomb measurements and investigated how temperature and component characteristics influenced performance and structural integrity. This work gave me practical experience linking a compact physical model with detailed CFD, evaluating sensitivities, and checking predictions against measurements. It led to a peer-reviewed publication in *Journal of Energy and Power Technology*.

My thermal-modelling background also includes research at Karlsruhe Institute of Technology, where I applied the reaction–diffusion manifold method to premixed flame–wall interactions and validated reduced kinetic models against detailed kinetics. At TU Darmstadt, I used COMSOL finite-element models to study non-isothermal Couette flow and natural convection, comparing low-Mach-number and Boussinesq formulations. These projects strengthened my ability to select suitable physical approximations and assess the fidelity of reduced models.

At KTH Royal Institute of Technology, my doctoral research focuses on upper-airway aeroacoustics and fluid–structure interaction. I develop compressible LES–FEM simulations in STAR-CCM+, conduct mesh and time-step convergence studies, and compare analytical model predictions with COMSOL calculations. I use NAISS HPC resources at PDC and NSC and lead analysis, visualisation, and manuscript preparation. Peer-reviewed writing and collaboration across fluid mechanics, structural dynamics, and acoustics have given me experience explaining modelling methods, assumptions, and limitations clearly.

I hold an M.Sc. in Mechanics from TU Darmstadt, with a final grade of 1.33, and am scheduled to defend my Ph.D. thesis at KTH on 4 December 2026. I use English for scientific writing and research communication; my Swedish is currently A2.

My experience to date is in industrial transient modelling and academic CFD. I would need to develop hands-on proficiency in nuclear system codes and reactor-specific safety methods. I would bring a foundation in physical model development, code coupling, parameter studies, and verification and validation, together with a commitment to developing the specialist knowledge needed to contribute to your team.

I would welcome the opportunity to discuss how these strengths could support the Fluid Systems Engineering team and Blykalla's development of lead-cooled reactor technology.

Kind regards,

Peng Li